

RESTU ANGGORO KASIH

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Bachelor's degree in Mathematics with a strong foundation in data analysis, machine learning, and AI implementation. Experienced in managing projects involving data preprocessing, model development, hyperparameter tuning, performance evaluation, and requirement analysis. Proficient in statistical methods and programming, with a strong interest in applying data-driven approaches to problem-solving.

EDUCATION

YOGYAKARTA STATE UNIVERSITY

Bachelor of Science; Major in Mathematics

- Cum Laude with Cumulative GPA: 3.64/4.00

Sleman, IDN

September 2021 - January 2025

ORGANIZATION

Mathematics Student Association

2022

Staff, IT Division

- Performed maintenance and repairs on hardware inventory, including one computer, one router, and two printers, on a four-day rotation system to ensure optimal performance.
- Managed over 30 software installations and ensured the smooth operation of IT systems for all members.
- Researched, designed, and distributed 11 media publications in the form of videos and images related to the latest technological developments or interesting tech facts, coordinating with the communication and publication team.

Engineering and Technology Club (INFINITE)

2023

Staff, Competition Division

- Conducted research and monitoring of more than 15 IT-related competitions, gathering and distributing relevant information to members.
- Managed an archive of over 25 competition-related documents, including proposals and other documentation.

Student Executive Board

2023

Staff, Research, Data, and Digital Products

- Analyzed and visualized more than 10 survey datasets using Excel or Python to support strategic decision-making.
- Designed and produced over 10 informative infographics based on data analysis results, collaborated with the communication and publication team to ensure consistency and effectiveness in media content.

PROJECTS

Tea Leaf Disease Classification

- Developed an EfficientNetV2S-based image classification model utilizing transfer learning to identify and categorize six distinct tea leaf diseases, achieving 98.33% accuracy in disease detection on the validation dataset.
- Optimized model accuracy from 94.33% to 98.2% through hyperparameter tuning, data augmentation, and model architecture optimization.
- Applied advanced data preprocessing techniques (image normalization, contrast enhancement, and augmentation) on 1,500 images, improving feature extraction and model generalization.
- Evaluated model performance with 98.33% accuracy, 0.1169 loss, 98.33% precision, 98.36% recall, 98.33% F1-score, and AUC: 0.9976, ensuring high classification reliability.

Land Use Land Cover (LULC) Classification & Web Deployment

- Designed an LULC classification system using Google Earth Engine (GEE) and Python, processing Sentinel-2 imagery with CNN, achieving 90.31% accuracy and Cohen's Kappa 85.49%.
- Applied geospatial feature extraction, integrating 6 spectral indices (EVI, NBR, NDMI, NDWI, NDBI, NDBaI) and elevation data to enhance classification performance.
- Built and deployed a Flask-based REST API and a React.js frontend, improving accessibility and scalability for geospatial data analysis.

- Enabled automatic area calculation for 4 land cover types (Water, Building, Plantation Forest, Herbaceous)

Customer Segmentation Using Birch Clustering

- Built a BIRCH clustering model to segment 2,000+ customers based on purchasing behavior and demographic data, improving customer targeting.
- Determined optimal cluster count using Elbow Method and evaluated segmentation quality with Silhouette Score (0.695) and Davies-Bouldin Index (0.440), ensuring well-separated and compact customer groups.
- Enhanced clustering efficiency by 40% through optimized feature selection, PCA, feature engineering, normalization, and handling of missing/duplicate values.
- Analyzed spending behavior across customer segments, identifying 3 key customer groups and deriving insights to optimize marketing strategies and improve campaign effectiveness.

Scraping-Based Data Collection and Classification

- Implemented a web scraping system using Google Play Scraper and BeautifulSoup to extract 4,000+ user reviews for sentiment analysis, applying text preprocessing techniques such as regular expressions, tokenization, stopwords removal, stemming, and slang word normalization to improve data quality.
- Generated actionable insights from sentiment analysis by visualizing trends using WordCloud and Seaborn, identifying 3 dominant sentiment categories that influenced model interpretation.
- Performed sentiment classification using multiple machine learning models, applied SMOTE to address class imbalance, performed hyperparameter tuning, and achieved 96.92% accuracy using SVM.

AWARDS

- **3rd Place** – LIKMI, Applied Technology Business Category, *Yogyakarta State University* 2023
Project: "FO-Shion (Find Your Shade Cushion): A Spark AR Studio-Based Solution for Choosing the Right Cushion Shade for Skin Tone".
- **Finalist** – TrackAML Hackathon *PPATK* 2023
Project: "A Combined Fraud Detection and Ensemble Learning System for AML and Terrorism Indicators in Anomaly Data Analysis".
- **Finalist** – Abdidaya Ormawa, PPK Ormawa, Ministry of Education, Culture, Research, and Technology 2023
Project: Community service to build a climate-resilient village through agroforestry, biogas, biopore infiltration systems, and MSME empowerment using agroforestry products like eco-printing.

ADDITIONAL

Technical Skills: Python, SQL, Tableau, MATLAB, JavaScript, R, HTML/CSS, Machine Learning, Deep Learning, Data Visualization, Web Scraping, Microsoft Office, Flask API.

Languages: Native in Indonesian; Conversational Proficiency in English.

Certifications & Training: Fundamentals of AI, Data Visualization, SQL, Python, Web Programming, JavaScript, Front-End Web – Dicoding, 2024.